

Statistik_Probetest_2026

- (1) A plumbing contractor obtains 60% of her boiler circulators from a company whose defect rate is 0.5%, and the rest from a company whose defect rate is 1%. What is the probability that a randomly chosen circulator is defective? If the chosen circulator is defective, what is the probability that it came from the first company?
- a. 0.007 and 0.571
 - b. 0.034 and 0.882
 - c. 0.034 and 0.118
 - d. 0.007 and 0.429**

-
- $P(A) = 0.60$ (60% kommen von Firma 1)
 - $P(B) = 0.40$ (Der Rest, also $100\% - 60\% = 40\%$)
 - $P(D|A) = 0.005$ (Defektrate Firma 1 ist 0.5%)
 - $P(D|B) = 0.01$ (Defektrate Firma 2 ist 1%)

$$P(D) = P(A) \cdot P(D|A) + P(B) \cdot P(D|B)$$

$$P(D) = (0.60 \cdot 0.005) + (0.40 \cdot 0.01) = 0.007$$

$$P(A|D) = \frac{P(D|A) \cdot P(A)}{P(D)} = \frac{0.005 \cdot 0.6}{0.007} = 0.4287$$

(2) Can the function

$$p(x) = \begin{cases} ax^2 + 2x - 1, & x = 1, 2, 3 \\ 0, & \text{else} \end{cases}$$

with $a \in \mathbb{R}$ be the probability mass function for a discrete random variable?

- a. Yes, only for a unique *positive* a .
 - b. Yes, only for a unique *negative* a .
 - c. No, because probabilities can *not* be greater than 1.
 - d.** No, because probabilities can *not* be negative.
-

$$\begin{aligned} & ax^2 + 2x - 1 \\ a + 2 - 1 + 4a + 4 - 1 + 9a + 6 - 1 &= 1 \\ \implies 14a + 8 &= 0 \\ \implies 14a &= -8 \\ \implies a &= -\frac{8}{14} \end{aligned}$$

Allerdings wenn man das dann einsetzt:

$$\begin{aligned} & -\frac{8}{14} \cdot x^2 + 2x - 1 \\ -\frac{8}{14} \cdot 3^2 + 3 \cdot 2 - 1 &= -0.1428 \end{aligned}$$

- (3) Pumpkins grown on a farm have normally distributed weights with a standard deviation of 2 kg. What is the approximate expected weight if 84% of the pumpkins weigh less than 16 kg?
- a. ≈ 12
 - b. ≈ 14**
 - c. ≈ 15
 - d. ≈ 11
-

$$P(X < 16) = 0.84$$

Standardisierung:

$$Z = \frac{x - \mu}{\sigma} \implies 0.7995 = \frac{16 - \mu}{2} \implies 1.599 = 16 - \mu \implies 14.401 = \mu$$

(4) Given a random sample $X_1, \dots, X_{64}$ from a distribution with expectation 1.2 and variance 4. Let

$$\bar{X} = \frac{1}{64} \sum_{i=1}^{64} X_i$$

be the sample mean. Which of the following R-functions approximates the value of $P(\bar{X} \leq 1.55)$ based on the Central Limit Theorem.

- a. `pnorm(0.175)`
 - b. `pnorm(1.55, 1.2, 4)`
 - c. `pnorm(1.4)`**
 - d. `pnorm(0.35, 0, 0.5)`
-

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{\sqrt{4}}{\sqrt{64}} = \frac{2}{8}$$

$$z = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}} = \frac{1.55 - 1.2}{\frac{2}{8}} = 1.4$$

$$\Rightarrow P(Z \leq 1.4)$$

(5) Let X be a random variable with density

$$f_X(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{else} \end{cases}.$$

Define $Z = -\ln X$, what is the third quartile of Z ?

The following table of approximate values of the natural logarithm is useful

x	1	2	3	4	5	...
$\approx \ln x$	0	0.7	1.1	1.4	1.6	...

- a. Between $1/2$ and 1
- b. Smaller than $1/2$
- c. is not unique
- d.** Bigger than 1

Gesucht ist

$$P(Z \leq q) = 0.75$$

und da $Z = -\ln(X)$

$$P(-\ln X \leq q) = 0.75$$

$$P(\ln X \geq -q) = 0.75$$

$$P(X \geq e^{-q}) = 0.75$$

Setzen wir das gleich 0.75:

$$1 - e^{-q} = 0.75$$

Nach q auflösen

$$e^{-q} = 1 - 0.75$$

$$e^{-q} = 0.25$$

Wir wissen, dass $0.25 = \frac{1}{4}$.

$$e^{-q} = \frac{1}{4}$$

Logarithmieren wir beide Seiten:

$$-q = \ln\left(\frac{1}{4}\right)$$

Nach den Logarithmusgesetzen gilt $\ln(1/a) = -\ln(a)$. Also:

$$-q = -\ln(4)$$

$$q = \ln(4)$$

Schauen wir in die Tabelle aus der Aufgabenstellung:

- $\ln 4 \approx 1.4$

(6) Which of the following holds **true** for two independent random variables X and Y with finite variance?

- a. $\text{Var}(2X - Y) \leq 4 \text{Var}(X)$
- b. $\text{Var}(X + Y) \leq \text{Var}(Y)$
- c.** $\text{Var}(X + Y) \geq \text{Var}(X)$
- d. $\text{Var}(2X - Y) = 2 \text{Var}(X) - \text{Var}(Y)$

- (7) In the situation of a left-sided one-sample t -test we find $\bar{x} = -8$, $s = 5$ and $n = 25$. For a given significance level we find the rejection region $R = (-\infty, -2.2]$. Then, for the null hypothesis $H_0 : \mu = -5$ holds
- a. we reject H_0 , and we would also reject for any smaller significance level.
 - b. we do not reject H_0 , but we would reject if the significance level was chosen small enough.
 - c.** we reject H_0 , and we would also reject for any larger significance level.
 - d. we do not reject H_0 , but we would reject if the significance level was chosen large enough.

$$T = \frac{x - \mu_0}{s/\sqrt{n}} = \frac{-8 + 5}{\frac{5}{\sqrt{25}}} = -3$$

(8) You perform a χ^2 -test for independence in R using `chisq.test()`.

```
x <- c(...)
y <- c(...)
chisq.test(rbind(x, y), correct = FALSE)
# Pearson's Chi-squared test
#
# data:  rbind(x, y)
# X-squared = 3.05, df = 1, p-value = 0.08074
```

What is the number of cells (rows $\times$ columns) in the two-way table `rbind(x, y)`?

- ☒ a. 4
- ☐ b. 5
- ☐ c. 6
- ☐ d. 2

$$df = (i - 1) * (j - 1) = 1$$

$$\implies i, j = 2 \implies 2 \cdot 2 = 4$$

(9) In general, how does doubling the sample size changes the two-sided confidence interval size?

- ☐ a. Halves the interval size.
- ☐ b. Multiplies the interval size by $\sqrt{2}$.
- ☒ c. Divides the interval size by $\sqrt{2}$.
- ☐ d. Double the interval size.

$$x \pm t \left(n - 1, 1 - \frac{\alpha}{\sqrt{2}} \right) \cdot \frac{s}{\sqrt{n}}$$

(10) What is the critical t -value for a two-sided 90% confidence interval estimate from a sample of size 15 using R?

- ☐ a. `qt(0.95, df = 15)` ≈ 1.753
- ☐ b. `qt(0.9, df = 15)` ≈ 1.351
- ☒ c. `qt(0.95, df = 14)` ≈ 1.761
- ☐ d. `qt(0.9, df = 14)` ≈ 1.345

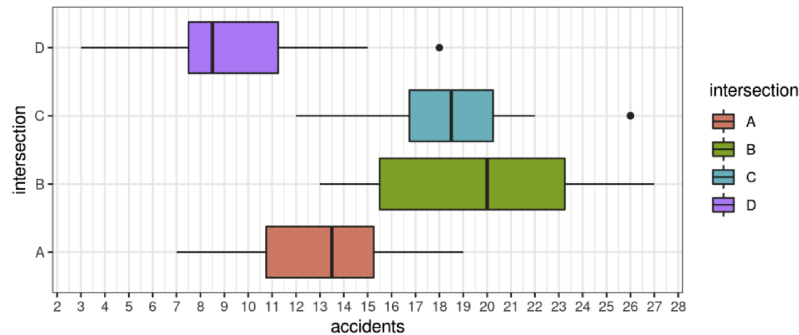
(11) A study cross-classified 1500 randomly selected adults in three categories of education level (not a high school graduate, high school graduate, and college graduate) and five categories of major sports interest (baseball, basketball, football, hockey, and tennis). If the p -value of an appropriate χ^2 -test for independence is 0.083, what can be said?

- ☒ a. There is sufficient evidence at the 10% significance level, but not at the 5% significance level, of a relationship between education level and sports interest.
- ☐ b. The data prove there is a relationship between education level and sports interest.
- ☐ c. There is sufficient evidence at the 5% significance level of a relationship between education level and sports interest.
- ☐ d. The p -value is greater than 0.10, so there is no evidence of a relationship between education level and sports interest.

(12) Which one of the following is a true statement?

- a. Statistics from smaller samples have less variability.
- b. Sample parameters are used to make inferences about population statistics.
- c. The larger the sample, the larger the spread in the sampling distribution.
- d.** Provided that the population size is significantly greater than the sample size, the spread of the sampling distribution does not depend on the population size.

(13) Data on the number of yearly accidents were collected from four intersections (A–D) over a 20 year period. Which of the following statements is **false**?



- a.** During at least 75% of years, intersection D had fewer accidents than the lowest 25% of years at intersection A.
- b. All of the accidents totals at intersection D were lower than the median number of accidents at intersection B.
- c. The maximum number accidents that occurred in a single intersection is 27.
- d. During at least 15 years, fewer than 12 accidents occurred at intersection D.

- (14) Suppose you do five independent right-sided tests for testing $H_0 : \mu = 38$, each at the $\alpha = 1\%$ significance level. What is the probability of committing a Type I error and incorrectly rejecting a true null hypothesis in at least one of the five tests?
- a. 1%
 - b. 95%
 - c. 22.6%
 - d. 4.9%**

Für einen Test ist Wahrscheinlichkeit 1% genau wie α

Für mehrere Tests

$$\begin{aligned}
 \text{Binomial : } B(5, \alpha) &\implies P(\text{one type 1 error}) = 1 - P(\text{no t1 error}) \\
 &= 1 - P(X = 0) \\
 &= 1 - \left(\binom{5}{0} \cdot \alpha^0 \cdot (1 - \alpha)^{5-0} \right) \\
 &= 1 - (1 \cdot 1 \cdot (1 - \alpha)^5) \\
 &= 1 - (1 - 0.01)^5 \\
 &= 0.049
 \end{aligned}$$

- (15) Suppose the correlation of two random variables X, Y is negative. Given two points (x_1, y_1) and (x_2, y_2) from the scatterplot, which of the following is possible?
- I. $x_1 > x_2$ and $y_1 < y_2$.
 - II. $x_1 > x_2$ and $y_1 > y_2$.
 - III. $x_1 < x_2$ and $y_1 > y_2$.
- ☒ a. I, II and III
 - ☐ b. Only I
 - ☐ c. I and II
 - ☐ d. Only II

Wert	Korrelationstyp	Bedeutung
$r \approx 1$	Stark positiv	Variablen steigen/sinken gemeinsam (gleiche Richtung).
$r \approx 0$	Keine (Nullkorrelation)	Kein linearer Zusammenhang erkennbar.
$r \approx -1$	Stark negativ	Eine Variable steigt, die andere sinkt (Gegenbewegung).

Allerdings gibt es ausreißer

- (16) Suppose the average score on a certain test is 500 with a standard deviation of 100. If each score is increased by 25, what are the new mean and standard deviation?
- ☒ a. Mean: 525, Standard deviation: 100
 - ☐ b. Mean: 500, Standard deviation: 125
 - ☐ c. Mean: 500, Standard deviation: 100
 - ☐ d. Mean: 525, Standard deviation: 125

$$\text{slope} = \frac{s_{xy}}{s_{xx}} =$$

- (17) In a simple linear regression model $y_i = ax_i + b + \epsilon_i$ the parameters are estimated via least squares. For the mean and the empirical variance of the x and y values we obtain $\bar{x} = 5$, $s_x^2 = 4$, $\bar{y} = 7$ and $s_y^2 = 9$. It holds that
- ☐ a. the regression line goes through $(7, 7)$.
 - ☐ b. the slope of the regression line is larger than 1.5.
 - ☒ c. the slope of the regression line is smaller than or equals 1.5.
 - ☐ d. the regression line goes through $(5, 6)$.

$$a = r \cdot \frac{s_y}{s_x} = r \cdot 1.5$$

(18) The sampling distribution of the sample mean is close to the normal distribution ...

- a. ... if n is large, and the population has finite variance.
- b. ... if the standard deviation of the population is known.
- c. ... no matter what the value of n or what the distribution of the population.
- d. ... only if both the population has a normal distribution and n is large.

-> CLT

(19) In the context of a statistical test at significance level α , the p -value is below α . Which of the following is **true**?

- a. The test statistic lies in the rejection region.
- b. The rejection region would be bigger if we would have used level $\alpha/2$.
- c. The value of the null hypothesis lies within the $(1 - \alpha)$ confidence interval computed from the data.
- d. We would also reject at level $\alpha/2$.

(20) A 95% confidence interval for a population mean based on a sample of size 400 was (20, 28). Which of the following is true?

- a. Over many samples, 95% of sample means should lie within an interval made by this method.
- b. If we were to collect a new sample, then the interval created using it would contain the true population mean with probability 95%.
- c. There is a 95% probability that the true value of the population mean is 24.
- d. The sample mean has a 95% chance of being in (20, 28).